

Determinants and quadratic forms

Determinant rules

- The determinant of a triangular (square!) matrix, $\det \mathbf{A}$, is the product of the entries on its diagonal
- In getting the matrix into triangular form using Gaussian elimination:
 - If two rows are exchanged the determinant is multiplied by -1
 - If a multiple of one row is subtracted from another the determinant is unchanged
- A square matrix is singular if and only if its determinant is zero
- $\det(\lambda \mathbf{A}) = \lambda^n \det \mathbf{A}$
- $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$
- $\det \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}$

The 2×2 case

Let

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

We write

$$\det \mathbf{A} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Reducing via Gaussian elimination we get

$$\begin{bmatrix} a & b \\ 0 & d - b\frac{c}{a} \end{bmatrix}$$

The product of the diagonal entries is

$$\det \mathbf{A} = ad - ab\frac{c}{a} = ad - bc$$

A longer example (let k keep track of # times rows switch)

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 3 & 7 \\ 5 & 8 & 6 \end{bmatrix}$$

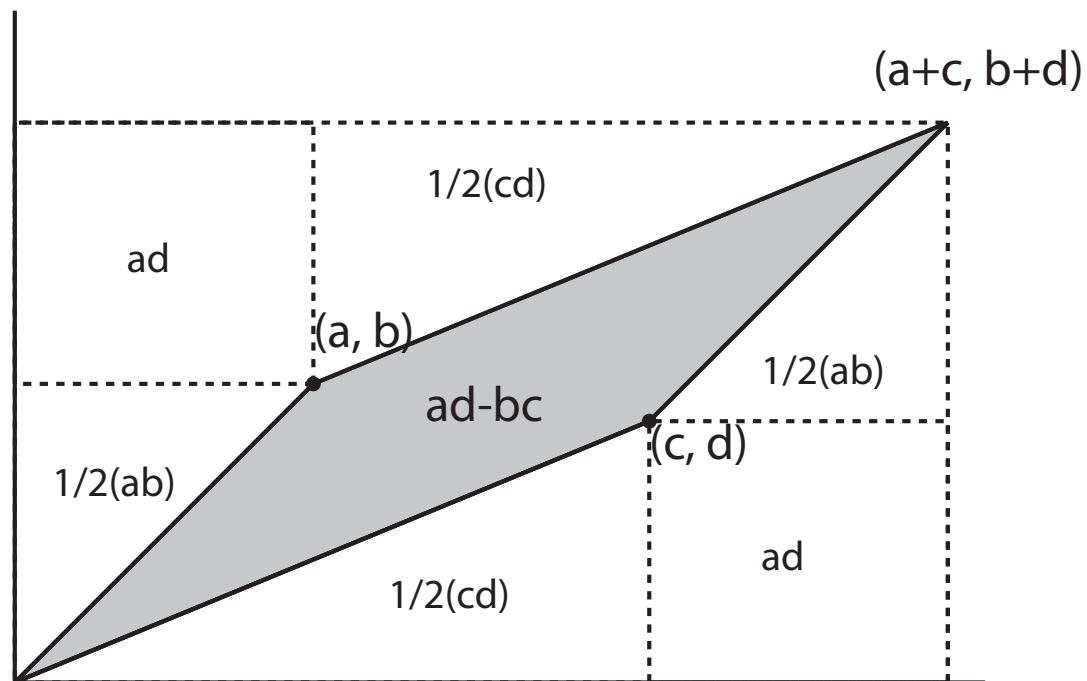
$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & 0 & 3 \\ 0 & \frac{1}{2} & -4 \end{bmatrix} \text{ and } k = 0$$

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & \frac{1}{2} & -4 \\ 0 & 0 & 3 \end{bmatrix} \text{ and } k = 1$$

$$\det \mathbf{A} = (-1)^k (3) = -3$$

What is a determinant?

For 2×2 matrices they have a natural interpretation:



The area of the parallelogram is

$$|(a+c)(b+d) - 2ad - ab - cd| = |ab + ad + bc + cd - 2ad - ab - cd| = |ad - bc|$$

Transposition

The transpose of a matrix, $\mathbf{A}^T$ is the matrix whose rows are the columns of $\mathbf{A}$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 0 & 1 \\ 5 & 5 & 5 & 5 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 5 \\ 2 & 1 & 5 \\ 3 & 0 & 5 \\ 4 & 1 & 5 \end{bmatrix}$$

Last class we discussed how the rank of a matrix could be computed by looking at the maximal number of linearly independent *rows* or *columns*

This implies that

$$\text{rank } \mathbf{A}^T = \text{rank } \mathbf{A}$$

We can get more than this:

$$\det \mathbf{A}^T = \det \mathbf{A}$$

(For the 2×2 case, this is obvious)

Inner products

If $\mathbf{x}$ and $\mathbf{y}$ are both n —vectors, then the product $\mathbf{x}^T \mathbf{y}$ is the scalar

$$x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

This is called the *inner product* or *dot product* or *scalar product* of $\mathbf{x}$ and $\mathbf{y}$ and is written

$$\mathbf{x}^T \mathbf{y} \text{ or } \mathbf{x} \cdot \mathbf{y} \text{ or } \langle \mathbf{x} \mathbf{y} \rangle$$

Example: Let $\mathbf{p}$ be a vector of prices of n goods and $\mathbf{x}$ be a vector of quantities of those n goods; then $\mathbf{x} \cdot \mathbf{p} = \sum_{i=1}^n p_i x_i$ is total expenditure

Length and distance

As $\mathbf{x} \cdot \mathbf{x} = \sum_i x_i^2$, it follows that the distance between the origin $\mathbf{0}$ and $\mathbf{x}$ is $\sqrt{\mathbf{x} \cdot \mathbf{x}}$

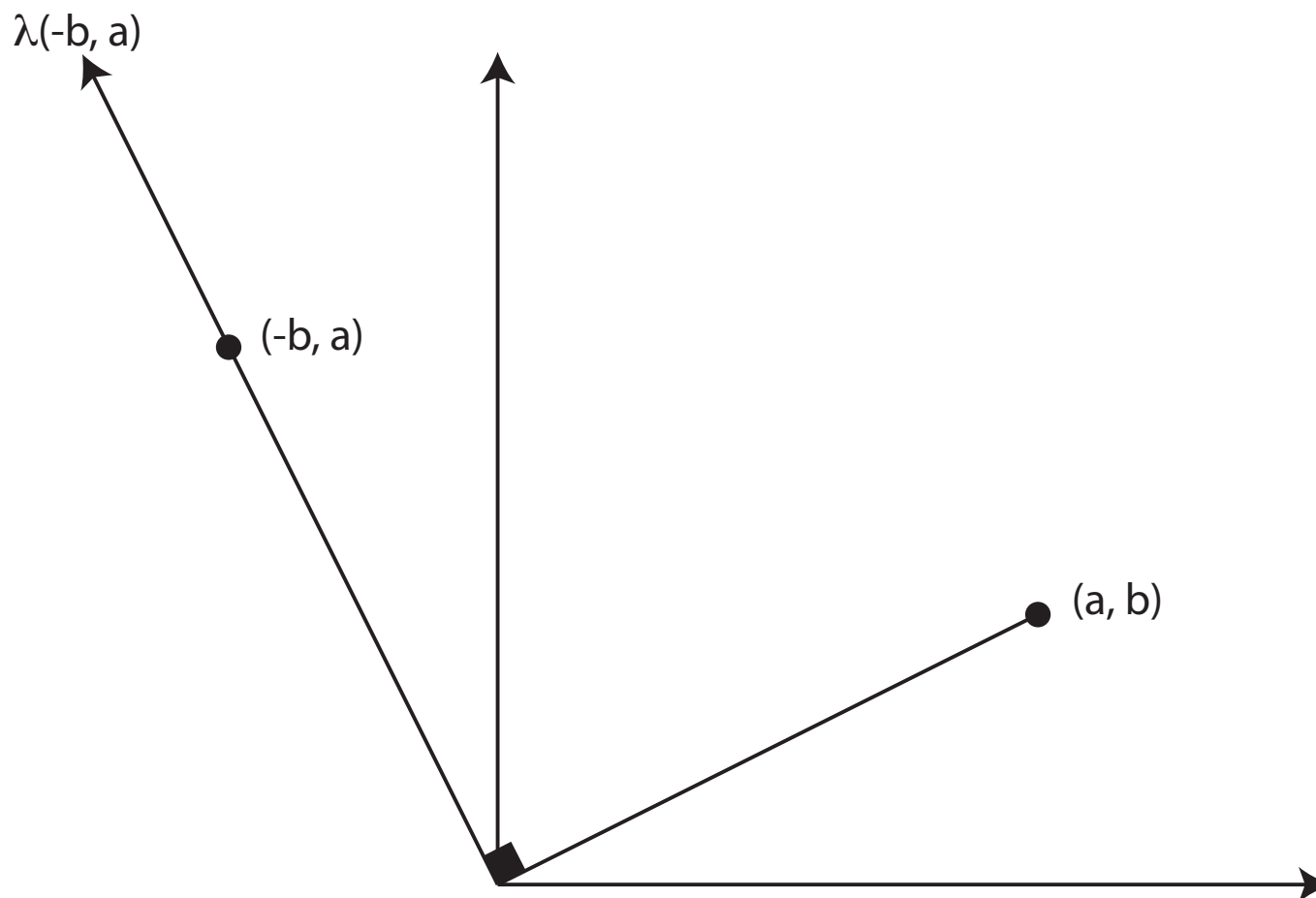
We call $\sqrt{\mathbf{x} \cdot \mathbf{x}}$ the *length*, *magnitude*, or *norm* of $\mathbf{x}$, and denote it $\|\mathbf{x}\|$

- $\|\mathbf{a}\| > 0$ if $\mathbf{a} \neq \mathbf{0}$
- $|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$ (Cauchy-Schwarz inequality)
- $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$ (Triangle inequality)

(We'll prove the latter two statements shortly)

Orthogonality

Let $\mathbf{a}$ and $\mathbf{b}$ be two n -vectors; then $\mathbf{a}$ is *orthogonal* to $\mathbf{b}$ if $\mathbf{a} \cdot \mathbf{b} = 0$



The geometry of dot products more generally

$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$

$\cos 90^\circ = 0$ and so $\mathbf{a} \cdot \mathbf{b} = 0$ if they're orthogonal

$\cos 0^\circ = 1$ and so $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\| \|\mathbf{a}\|$

$\cos \theta \in [-1, 1]$ for all angles θ

Using this, the Cauchy-Schwarz inequality is immediate:

$$|\mathbf{a} \cdot \mathbf{b}| = \|\mathbf{a}\| \|\mathbf{b}\| |\cos \theta| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

The triangle inequality follows from the Cauchy-Schwarz inequality:

We want to prove $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$

$$\begin{aligned}\|\mathbf{a} + \mathbf{b}\|^2 &= \langle \mathbf{a} + \mathbf{b}, \mathbf{a} + \mathbf{b} \rangle \\ &= \langle \mathbf{a}, \mathbf{a} \rangle + 2\langle \mathbf{a}, \mathbf{b} \rangle + \langle \mathbf{b}, \mathbf{b} \rangle \\ &\leq \langle \mathbf{a}, \mathbf{a} \rangle + 2|\langle \mathbf{a}, \mathbf{b} \rangle| + \langle \mathbf{b}, \mathbf{b} \rangle \\ &\leq \langle \mathbf{a}, \mathbf{a} \rangle + 2\|\mathbf{a}\|\|\mathbf{b}\| + \langle \mathbf{b}, \mathbf{b} \rangle \quad (\text{Cauchy-Schwarz}) \\ &= \|\mathbf{a}\|^2 + 2\|\mathbf{a}\|\|\mathbf{b}\| + \|\mathbf{b}\|^2 \\ &= (\|\mathbf{a}\| + \|\mathbf{b}\|)^2\end{aligned}$$

Taking a square root on each side we get

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\| \quad \square$$

Quadratic forms and symmetric matrices

- A *quadratic form* is a homogenous quadratic polynomial in n variables
 - A *homogeneous polynomial* is a polynomial whose nonzero terms all have the same degree: in words, the sum of the exponents of each term is constant
 - A *homogenous quadratic polynomial* is a polynomial whose terms all have degree 2
 - ax^2
 - $ax^2 + bxy + cy^2$
 - $ax^2 + by^2 + cz^2 + dxy + exz + fyz$ (with coefficients a, b, c, d, e, f)

Consider the two variable quadratic form

$$q(x, y) = ax^2 + bxy + cy^2$$

Using our techniques from last class we know we can write

$$q(x, y) = \mathbf{z}^T \mathbf{A} \mathbf{z}$$

where

$$\mathbf{z} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } \mathbf{A} = \begin{bmatrix} a & s \\ t & c \end{bmatrix}$$

with $s + t = b$

If we impose the condition that $\mathbf{A}$ is symmetric (so that $\mathbf{A} = \mathbf{A}^T$, or $a_{ij} = a_{ji}$ for all i, j) then we have $s = t = \frac{b}{2}$

There is therefore a unique symmetric matrix $\mathbf{A}$ satisfying $q(x, y) = \mathbf{z}^T \mathbf{A} \mathbf{z}$

This generalizes:

Let $q(\mathbf{x})$ be a quadratic form in n variables, $x_1, \dots, x_n$

Then $q(\mathbf{x})$ is equivalent to the expression

$$\mathbf{x}^T \mathbf{A} \mathbf{x}$$

for a unique symmetric $n \times n$ matrix $\mathbf{A}$ and $\mathbf{x}^T = [x_1, x_2, \dots, x_n]$

Takeaway point: A symmetric $\mathbf{A}$ is uniquely determined by the coefficients in $q(\mathbf{x})$

Definite and semidefinite quadratic forms

For a given quadratic form $q(\mathbf{x})$ we can ask whether $q(\mathbf{x}) > 0$ (or less stringently, $q(\mathbf{x}) \geq 0$) for every non-zero $\mathbf{x}$

If $q(\mathbf{x}) > 0$ for every non-zero $\mathbf{x}$ then $q(\mathbf{x})$ is *positive definite*

- Example $q(\mathbf{x}) = x_1^2 + x_2^2 + \dots + x_n^2$

If $q(\mathbf{x}) \geq 0$ for every $\mathbf{x}$ then $q(\mathbf{x})$ is *positive semidefinite*

- Example $q(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 - 2x_2x_3$
 - $q(\mathbf{x}) = x_1^2 + (x_2 - x_3)^2 \geq 0$ but $q(0, 1, 1) = 0$

More on positive (and negative) definiteness

If $q(\mathbf{x}) < 0$ for every non-zero $\mathbf{x}$ then $q(\mathbf{x})$ is *negative definite*

If $q(\mathbf{x}) \leq 0$ for every $\mathbf{x}$ then $q(\mathbf{x})$ is *negative semidefinite*

Most quadratic forms are neither positive nor negative semidefinite

Example: $q(x, y) = xy$

- $q(1, 1) > 0$
- $q(1, -1) < 0$

Testing quadratic forms

We can test if a quadratic form is positive (negative) (semi)definite by testing the associated symmetric matrix $\mathbf{A}$ that satisfies $q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$

If $\mathbf{A}$ is 2×2 , and recall, $\mathbf{A} = \begin{bmatrix} a & c \\ c & d \end{bmatrix}$, then:

- $\mathbf{A}$ is PD iff $a, d > 0$ and $\det \mathbf{A} > 0$
- $\mathbf{A}$ is PSD iff $a, d \geq 0$ and $\det \mathbf{A} \geq 0$
- $\mathbf{A}$ is ND iff $a, d < 0$ and $\det \mathbf{A} > 0$
- $\mathbf{A}$ is NSD iff $a, d \leq 0$ and $\det \mathbf{A} \geq 0$

Testing matrices for “definiteness” is something we do a lot of in optimization^a

Definitions

- A *submatrix* of $\mathbf{A}$ is a matrix obtained by deleting some rows and columns of $\mathbf{A}$
- A *principal submatrix* of a square $\mathbf{A}$ is a submatrix obtained by requiring that if the k th row is deleted then the k th column is deleted
- A *leading principal submatrix* of $\mathbf{A}$ is a principal submatrix obtained by deleting the last m rows and columns for some $m < n$

^aTaylor’s Theorem implies the second-order behavior of smooth functions of several variables can be described by the quadratic form of the function’s Hessian, which is a symmetric matrix of the function’s partial derivatives

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

The 7 principal submatrices are:

$$\begin{bmatrix} a \end{bmatrix} \begin{bmatrix} e \end{bmatrix} \begin{bmatrix} i \end{bmatrix} \begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} a & c \\ g & i \end{bmatrix} \begin{bmatrix} e & f \\ h & i \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

The 3 leading principal submatrices are:

$$\begin{bmatrix} a \end{bmatrix} \begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Minors

- A *minor* of $\mathbf{A}$ is the determinant of a submatrix of $\mathbf{A}$
- A *principal minor* of $\mathbf{A}$ is the determinant of a principal submatrix of $\mathbf{A}$
- A *leading principal minor* of $\mathbf{A}$ is the determinant of a leading principal submatrix of $\mathbf{A}$

Test for positive definiteness: An $n \times n$ symmetric matrix $\mathbf{A}$ is positive definite if and only if its leading principal minors are all positive

Test for positive semidefiniteness: An $n \times n$ symmetric matrix $\mathbf{A}$ is positive semidefinite if and only if its principal minors are all non-negative

(Test for negative (semi)definiteness by applying above conditions to $-\mathbf{A}$)

Least squares estimation

Suppose you have data consisting of 2 observations on three variables, (y, x_1, x_2) ; you want to find a function of the form

$$y = \beta_1 x_1 + \beta_2 x_2$$

that fits the data best: β_1 and β_2 minimize the expression

$$q(\beta_1, \beta_2) = \sum_{i=1}^2 (y_i - \beta_1 x_{i1} - \beta_2 x_{i2})^2$$

Let

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$$

Note that

$$\mathbf{Xb} = \begin{bmatrix} \beta_1 x_{11} + \beta_2 x_{12} \\ \beta_1 x_{21} + \beta_2 x_{22} \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \quad \mathbf{X}\mathbf{b} = \begin{bmatrix} \beta_1 x_{11} + \beta_2 x_{12} \\ \beta_1 x_{21} + \beta_2 x_{22} \end{bmatrix}$$

We can rewrite $q(\beta_1, \beta_2) = \sum_{i=1}^2 (y_i - \beta_1 x_{i1} - \beta_2 x_{i2})^2$ as

$$q(\mathbf{b}) = \langle \mathbf{y} - \mathbf{X}\mathbf{b}, \mathbf{y} - \mathbf{X}\mathbf{b} \rangle$$

$$q(\mathbf{b}) = \langle \mathbf{y} - \mathbf{Xb}, \mathbf{y} - \mathbf{Xb} \rangle$$

Finding $\mathbf{b} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$ to minimize q is an optimization problem in 2 variables

$$q(\mathbf{b}) = \langle \mathbf{y}, \mathbf{y} \rangle + \langle \mathbf{Xb}, \mathbf{Xb} \rangle - 2\langle \mathbf{y}, \mathbf{Xb} \rangle$$

$$\begin{aligned} q(\beta_1, \beta_2) &= y_1^2 + y_2^2 \\ &+ (\beta_1 x_{11} + \beta_2 x_{12})^2 + (\beta_1 x_{21} + \beta_2 x_{22})^2 \\ &- 2(y_1(\beta_1 x_{11} + \beta_2 x_{12})) - 2(y_2(\beta_1 x_{21} + \beta_2 x_{22})) \end{aligned}$$

$$q(\mathbf{b}) = \langle \mathbf{y}, \mathbf{y} \rangle + \langle \mathbf{Xb}, \mathbf{Xb} \rangle - 2\langle \mathbf{y}, \mathbf{Xb} \rangle$$

$$\begin{aligned} q(\beta_1, \beta_2) &= y_1^2 + y_2^2 \\ &+ (\beta_1 x_{11} + \beta_2 x_{12})^2 + (\beta_1 x_{21} + \beta_2 x_{22})^2 \\ &- 2(y_1(\beta_1 x_{11} + \beta_2 x_{12})) - 2(y_2(\beta_1 x_{21} + \beta_2 x_{22})) \end{aligned}$$

Differentiating with respect to β_1, β_2 :

$$\frac{\partial q}{\partial \beta_1} = 2(\beta_1 x_{11} + \beta_2 x_{12})x_{11} + 2(\beta_1 x_{21} + \beta_2 x_{22})x_{21} - 2(y_1 x_{11} + y_2 x_{21}) = 0$$

$$\frac{\partial q}{\partial \beta_2} = 2(\beta_1 x_{11} + \beta_2 x_{12})x_{12} + 2(\beta_1 x_{21} + \beta_2 x_{22})x_{22} - 2(y_1 x_{12} + y_2 x_{22}) = 0$$

$$\frac{\partial q}{\partial \beta_1} = 2(\beta_1 x_{11} + \beta_2 x_{12})x_{11} + 2(\beta_1 x_{21} + \beta_2 x_{22})x_{21} - 2(y_1 x_{11} + y_2 x_{21}) = 0$$

$$\frac{\partial q}{\partial \beta_2} = 2(\beta_1 x_{11} + \beta_2 x_{12})x_{12} + 2(\beta_1 x_{21} + \beta_2 x_{22})x_{22} - 2(y_1 x_{12} + y_2 x_{22}) = 0$$

Moving back to matrix notation we can write these equations as

$$\mathbf{X}^T \mathbf{X} \mathbf{b} = \mathbf{X}^T \mathbf{y}$$

Assuming $\mathbf{X}^T \mathbf{X}$ has full rank (is invertible), this is an equation of the form $\mathbf{A} \mathbf{x} = \mathbf{y}$; thus the least squares solution is:

$$\mathbf{b}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Playing with our result

$$\mathbf{b}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Country	Life expectancy	Per Cap GDP	Pop density
USA	78	48	85
Russia	70	17	21
Brazil	73	12	62

$$\mathbf{X} = \begin{bmatrix} 48 & 85 \\ 17 & 21 \\ 12 & 62 \end{bmatrix} \text{ and } \mathbf{y} = \begin{bmatrix} 78 \\ 70 \\ 73 \end{bmatrix}$$

If we calculate $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ using the above matrices we get

$$\mathbf{b}^* = \begin{bmatrix} 0.3128 \\ 0.9561 \end{bmatrix} \text{ and so Life expectancy} = .3128 \text{ GDP} + .9561 \text{ Density fits}$$

the data best, in the sense of minimizing

$$\sum_{i=1}^3 (\text{Life expectancy}_i - b_1^* \text{GDP}_i - b_2^* \text{density}_i)^2$$